

Cryptographic observability feature policies

Legitimate metadata is separated from target leakage and sensitive raw identifiers

Hard leakage and sensitive raw fields

label	attack category
scenario	source file
raw IP identity	exact timestamp
client ID	username/password
payload/topic strings	



removed before model training

audit

Full audited metadata

upper-bound metadata after hard leakage removal

Broker-side secure metadata

broker/flow attributes without raw content or credentials

Strict encrypted flow metadata

duration, packets, bytes, IAT, rates, burstiness, TCP flags

duration

pkt count

bytes

IAT

rates

burst

TCP flags

Policy logic

$F_{\text{main}} = X \text{ minus } H$ $F_{\text{cons}} = F_{\text{main}} \text{ minus } S$ $F_{\text{common}} = \text{semantic intersection}$

High single-feature predictiveness alone is not treated as leakage.